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Research paper

Solar energy utilization for underfloor heating system in residential buildings

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A R T I C L E I N F O

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A B S T R A C T

In this paper, the thermal performance of a solar energy system for heating a residential building is experimentally investigated. In this system, the solar thermal energy is supplied via a flat plate solar collector that provides the heated water into a small-scale chamber, that represents the residential building. The experimental setup is installed in Jouf University campus (Latitude: 29° 47′ 1.9" N, Longitude: 40° 2′ 42.2" E) located at Al-Jouf Province, Saudi Arabia. In this experiment, the hot water supplied from solar collector is mixed with water in the storage tank, then extracted from bottom of the tank to be circulated in a copper tube loop located underneath the floor tiles of the chamber and hence increases the chamber's air temperature. Two circulation pumps are installed in the system, one to circulate the water between solar collector and storage tank and the other to circulate the water between storage tank and chamber. Measurements of several parameters were conducted to evaluate the overall thermal efficiency of the system. The experimental results showed an average efficiency of 51 %, when operating the system for eight-hours duration in the winter season. Additionally, an economic assessment was carried out to confirm the system's feasibility. This assessment demonstrated that the constructed system is economically feasible, with an estimated benefit-cost ratio of 1.42.

1. Introduction

Households are one of the most energy-intensive sectors of the economy, accounting for nearly 30 % of global energy consumption (Ürge-Vorsatz et al., 2015; Abdelmaksoud and Khalil, 2015; W. Abdelmaksoud, 2013). Domestic hot water (DHW) accounts for almost a quarter of all energy consumed in houses (Binks et al., 2016). To address the DHW requirement, a variety of energy technologies have been proposed. The majority of these technologies are powered by fossil fuels, which significantly contribute to greenhouse gas emissions and have negative environmental consequences (Book, 2011). The integration of solar water heating (SWH) systems with traditional heating systems offers significant potential for reducing fossil fuel usage, pollution, and greenhouse gas emissions (Daghigh and Shafieian, 2016). The analysis in (Serban et al., 2016) showed that if SWH systems are used for residential applications, the annual energy savings will be around 71 %, and the reduction of greenhouse gas emissions into the environment will be 18.5 tonnes of CO_2 throughout the system's lifespan, with a discounted payback period of 6.8–8.6 years.

The energy demand for regions with dense and growing populations is highly desired, such as in the Middle East region. The region's annual average solar intensity is 2000–3200 kWh/m^2, indicating a strong potential for using solar energy technology (Bahrampour et al., 2017; Mas'ud et al., 2018; Abdelmaksoud et al., 2021). Moreover, the Middle Eastern countries benefit from enormous oil and gas resources, with the region's energy relying heavily on fossil fuels (Nematollahi et al., 2016), i.e., in 2018, renewable energies supplied less than 1 % of the region's energy consumption (Bayomi and Fernandez, 2019). In addition, because of its dry and hot environment, the Middle East is an ideal location for the construction of SWH systems. In this context, there has been an increase in investment in SWH systems in recent years, for example, 2 million USD in 2016, which is predicted to climb by 0.5 million USD by the end of 2024 (Raisul et al., 2013).

A SWH system can run for more than 25 years without requiring major maintenance, making it a feasible investment (Balaji et al., 2018). Because of the mentioned economic and environmental benefits, the use of SWH systems has increased rapidly in the last decade (Bracamonte et al., 2015; Lin et al., 2015). Flat plate solar water heating (FPSWH) systems are the earliest form of SWH system, which are widely utilized due to its simple structure and low cost. FPSWH systems have been shown to have low thermal efficiency, particularly in cold seasons, due to substantial heat loss and low convective heat transfer coefficient (Kalogirou, 2004). For instance, when the FPSWH system is used in

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Nomenclatures

DHW	domestic hot water
SWH	solar water heating
FPSWH	flat plate solar water heating
PCM	phase change materials
ANSI	American national standards institute
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
Q_u	useful energy gain, W
E_i	input energy, W
$\dot{m}_w$	water mass flow rate, kg/s
ΔT_w	water temperature difference, °C
$\dot{V}_w$	water volume flow rate, m^3/s
C_{pw}	water specific heat at constant pressure, kJ/kg °C
I_R	Intensity of solar radiation, W/m^2
A_C	collector surface area, m^2
P_{pumps}	pumping power, W
I	Electric current, Amp and Initial capital cost, SAR
V	Electric voltage, Volt
BCR	Benefit-cost ratio
SAR	Saudi Arabian Riyal
B	Benefits of the project, SAR
O & M	operating and maintenance cost

Greek symbols

η_o	overall system efficiency, %
ρ_w	water density, kg/m^3
ω	uncertainty percentage, %

winter, the thermal efficiency drops from 75 % in summer to 40 %. Cruz et al. (2020) investigated the thermal performance, environmental and economic justifications of a FPSWH system. The results revealed that when the system was operated with an electrical backup, the payback period was 4.5 years. The technical and environmental performance of a FPSWH system was evaluated in a similar study (Koroneos and Nanaki, 2012). The study illustrated a payback period of 5 years and approximately 4280 € can be saved in Greece by installing a solar system throughout its lifespan. Several research on the energy, economic, and environmental implications of FPSWH systems have been introduced in Ref (Rassamakin et al., 2013; Yang and Liu, 2011; Sokhansefat et al., 2018; García et al., 2019; Yılmaz, 2018; Chopra et al., 2018). A theoretical investigation on the economic feasibility of flat plate vs evacuated tube solar collectors was performed by Nájera-Trejoa et al. (2016). The results revealed that the flat plate system had a 9-year return on investment, whereas the evacuated tube system had an 11-year return on investment.

Pervious investigations have been conducted on the aim of efficiency improvement for flat plate collectors. For instance, Kizildag et al. (2022) investigated the performance of three flat plate collector prototypes with transparent insulating materials. Their performance was compared to a conventional flat plate collector. The energy collected was found to be 1.4 and 2.5 times higher than the conventional collector in spring and winter seasons. Hashim et al. (2018a) experimentally investigated the effect of water flow rate on the outlet temperature and effectiveness of flat plate collector. The experimental results showed that with water flow rates of 5.3 and 6.51 L/min, the maximum temperatures were 51.4 °C and 49 °C, respectively. Ebaid et al. (2021) compared the performance of hemi-spherical solar collector against flat plate solar collector. The maximum efficiency of the hemi-spherical solar collector was found to be 69 %, compared to 42 % for the flat plate solar collector. A solar reflector was used along with a solar collector to boost the collector's reflectivity (Bhowmik and Amin, 2017). The experimental results showed a 10 % increase in the collector efficiency by using a reflector in the system. Shojaeizadeh et al. (2014) studied the impact of changing the concentration of propylene glycol in a propylene glycol–water solution on the efficiency of a flat plate collector. The experimental results showed that increasing the propylene glycol concentration from 25 % to 75 % improves the flat plate collector's efficiency and increasing the concentration from 75 % to 100 % decreases the maximum efficiency of the collector by 8.3 %.

Modern space heating technologies have been implemented in previous research and evaluated in terms of energy efficiency and economic feasibility. For instance, Faraj et al. (2022) conducted comparative energy and economic studies of underfloor hydronic heating system performance in terms of weather conditions, phase change materials (PCM) application, PCM type and characteristics, and PCM plate position at underfloor. The results demonstrated significant energy savings (169 USD annual saving and 3.94 years' payback period) when using coconut oil PCM in underfloor of the constructed system. Devaux and Farid (2017) conducted theoretical and experimental study for incorporating the PCM in walls, ceiling, and underfloor for space heating in winter season. A ten-day period was examined, revealing cost and energy savings of 42 % and32 %, respectively. Arsalan et al. (2023) utilized a hybrid solar heating system consisting of evacuated tube collectors, flat plate collectors, and convective radiator for a residential space heating. The energy efficiency analysis showed a 61 % maximum system efficiency and the economic analysis showed 0.82–1.6 benefit-cost ratio and 8.9–7.41 payback period.

The literature review shows that further research is necessary to evaluate (in terms of energy efficiency and economic feasibility) the utilization of flat plate solar collectors in heating the residential buildings. In this paper, a flat plate solar collector is utilized to provide a heated water into a small-scale chamber, that represents a residential building. A storage tank is installed between the solar collector and the chamber. The hot water supplied to the chamber, is extracted from the storage tank and circulated in a copper tube loop located underneath the floor tiles of the chamber and hence increase the chamber's air temperature. Several measured parameters were conducted to evaluate the thermal performance and economic feasibility of the constructed system. All these parameters were conducted on a 15-minutes interval for an 8-hours period in January 2022. Details for the experimental methodology and results are described the next sections.

2. Experimental

2.1. Methodology

An experimental study was conducted on an underfloor heating system that utilizes the solar thermal energy via a flat plate solar collector. The experiment consists of a small-scale chamber (represents a residential building), water storage tank, flat plate collector, and two circulation pumps. In this system, water is the working fluid enters the flat plate collector, receives the absorbed energy, and leaves the collector at a higher temperature. This hot water directly mixes with the water inside the storage tank to raise the average water temperature inside the tank. A pump is installed in the line between the collector and the tank to improve the heat transfer and keeps a homogeneous water temperature inside the tank. The chamber receives hot water from the storage tank via a valve located at the bottom of the tank. The other pump is installed in the line between the tank and the chamber to extract the high temperature water from the tank and circulates it inside the copper tube loop (located in underfloor of the chamber) to heat the chamber's air and returns the low temperature water to the tank.

The chamber dimensions are $1.2 \times 1.2 \times 1.5\, m^3$ and is thermally insulated from its surroundings. The water storage tank capacity is 150 liters of water and is used to hold a homogenous and hot water to be supplied to the chamber. The flat plate collector surface area is $1.5\, m^2$ and is used to absorb the solar radiation and converts it to heat, which is then absorbed by a circulated water in the collector. The hot water is circulated in a copper tube arranged as loop square pattern and embedded in a regular sand, located underneath the chamber's floor tiles. The heat transfers from the hot water in the tube, to the tube's surface, to the sand, to the floor tiles, and eventually to the chamber's air. Two circulation pumps, 370 W each, are installed to circulate the water in the system. One of the installed pump is used to circulate the water between the flat plate collector and storage tank. The other, is used to circulate the hot water between the storage tank and chamber. The flow diagram of the constructed system is shown in Fig. 1 and the detailed steps of constructing the whole system are shown in Fig. 2. Information about the tube loop configuration (single serpentine configuration in Fig. 2-b) are listed in Table 1. The system was constructed and instrumented to measure the chamber's average air temperature, chamber's inlet and outlet water temperatures, water flow rate, incident solar radiation, wind speed, and pumping power. The chamber's average air temperature measurement was conducted via four thermocouples installed in center of the chamber (see Fig. 2-f). The water temperature measurements were conducted via two thermocouples installed at the inlet and outlet of the copper tube loop located in the chamber's underfloor. Both water and air temperature measurements were conducted via HT-9815 Digital K Type Thermometers with ±2 % accuracy. The water flow rate measurement in the copper tube loop was conducted via a K24 Digital Turbine Flow Meter with accuracy ±1 % accuracy. The solar radiation intensity measurement was conducted via a digital solar power meter of type SM206-SOLAR with ±5 % accuracy. The wind speed measurement was conducted via a velocity meter of type UT363 Mini-digital Anemometer with ±5 % accuracy. The pumping power measurement was conducted via the measurements of both current and voltage through a DCM266 Digital Clamp Meter with ±2 % accuracy in current and ±1 % accuracy in voltage. All these measured quantities were conducted every 15-minutes interval for an 8-hours period (starting from 8:00 AM to 4:00 PM) in January 2022. However, before conducting this study, shakedown experiments were carried out to ensure repeatability of the experimental data.

$$\omega_{\eta_o} = \left[\left(\frac{\partial \eta_o}{\partial V_w^{\bullet}} \omega_{V_w^{\bullet}} \right)^2 + \left(\frac{\partial \eta_o}{\partial \Delta T_w} \omega_{\Delta T_w} \right)^2 + \left(\frac{\partial \eta_o}{\partial I_R} \omega_{I_R} \right)^2 + \left(\frac{\partial \eta_o}{\partial I} \omega_I \right)^2 + \left(\frac{\partial \eta_o}{\partial V} \omega_V \right)^2 \right]^{1/2} \quad (1)$$

2.2. Uncertainty

The measuring devices and their uncertainty percentage were described in the previous section and are listed in Table 2 as well. Based on the primary data conducted from these devices, the uncertainty percentage (ω) of the overall system efficiency (η_o) was estimated according to the mathematical correlations provided in Holman (2012). According to Eq. (7) (described in next section), η_o is function of the independent variables of water temperature difference (ΔT_w), water flow rate ($V_w^{\bullet}$), intensity of solar radiation (I_R), electric current (I), and electric voltage (V). Thus, the uncertainty percentage of overall efficiency ($\omega_{\eta o}$) was estimated according to the following correlation:

The equation above results in a 2.73 % uncertainty percentage of the overall system efficiency.

3. Governing equations

This section is devoted to describe the mathematical relations that were required to estimate the overall system efficiency. These relations were used under the assumption of constant thermo-physical properties of water, steady state conditions and neglected heat loss from the chamber. The overall system efficiency can be expressed as (Hashim et al., 2018b):

$$\eta_o = \frac{Q_u}{E_i} \quad (2)$$

where Q_u is the useful energy gain (W) and E_i is the input energy (W) to the system. The Q_u was calculated using the heat energy equation for the water across the chamber control volume as following:

$$Q_u = m_w^{\bullet} \; C_{pw} \; \Delta T_w \quad (3)$$

$$m_w^{\bullet} = \rho_w V_w^{\bullet} \quad (4)$$

where $m_w^{\bullet}$ is the water mass flow rate (kg/s), ΔT_w is the water temperature difference across the chamber control volume (°C), $V_w^{\bullet}$ is the water volume flow rate (m^3/s), C_{pw} and ρ_w are the water specific heat at constant pressure (kJ/kg °C) and water density (kg/m^3), respectively. The C_{pw} and ρ_w were assumed to be constant values, 4180 kJ/kg °C and

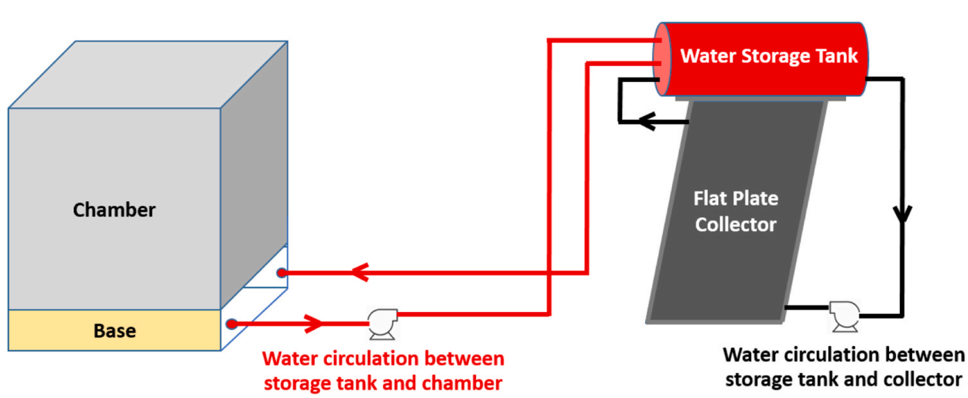

Fig. 1. Flow diagram of the constructed system.

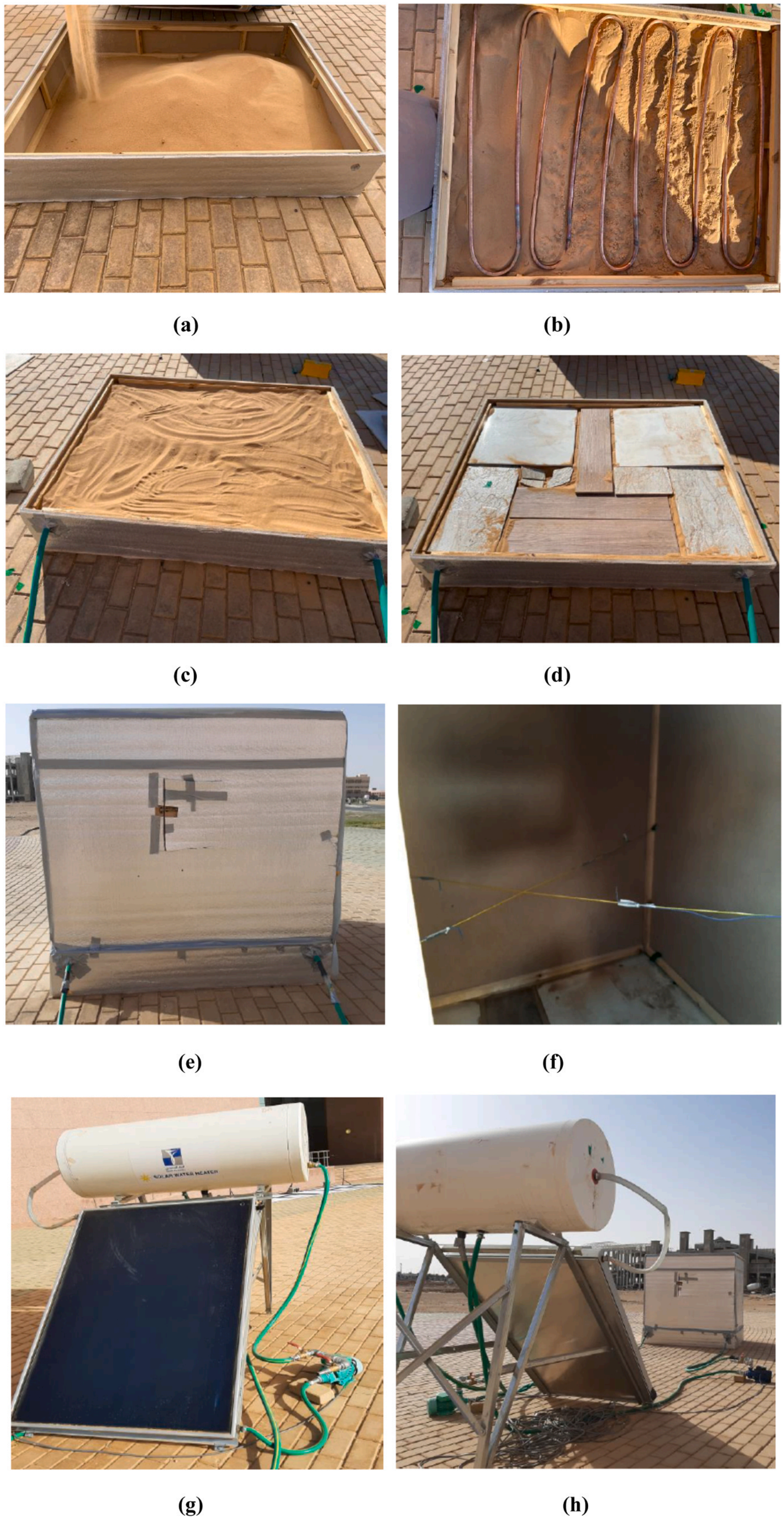

Fig. 2. Photographs of the constructed system: (a) filling the chamber's base with a sand layer, (b) inserting the copper tube loop in middle of the sand layer, (c) covering the copper tube with sand, (d) installing the floor tiles on the sand layer, (e) the constructed chamber, (f) four thermocouples installed inside the chamber, (g) flat plate solar collector and storage tank, and (h) final shape of the constructed system.

Table 1
Information about the tube loop configuration.

Item	Value
Tube inner diameter	0.015 m
Total length of the tube	12 m
Tube pitch	0.15 m
Water flow rate	26.5 L/min

Table 2
Uncertainty of the studied parameters.

Parameter	Uncertainty (ω)
Water temperature (°C)	±2 %
Air temperature (°C)	±2 %
Water flow rate (L/min)	±1 %
Solar radiation intensity (W/m^2)	±5 %
Wind speed (m/s)	±5 %
AC current (A)	±2 %
AC voltage (V)	±1 %
Overall system efficiency (%)	±2.73 %

1000 kg/m^3, respectively (Arora, 2009). Two sources of energy inputs (E_i) to the system, one is the incident solar radiation received by the solar collector and the other is the pumping power (two identical pumps) used to circulate water in the system. Thus, E_i can be expressed as following:

$$E_i = I_R \quad A_C + P_{pumps} \tag{5}$$

$P_{pumps} = 2IV$ (6) where I_R is the intensity of solar radiation (W/m^2), A_C is the collector surface area (m^2), P_{pumps} is the pumping power (W), I is the electric current (Amp), and V is the electric voltage (Volt). Based on the above mentioned equations, the overall system efficiency was estimated according to the following expression:

$$\eta_o = \frac{m^{\bullet}_w \quad C_{pw} \quad \Delta T_w}{I_R \quad A_C \quad + \quad P_{pumps}} \tag{7}$$

4. Results and discussions

This system was constructed to investigate the performance of utilizing the solar heated water in residential buildings for heating purpose during the winter seasons. The solar heated water was used to heat the air inside a test chamber that represents a residential building. The experiment was carried out in January 2022 in the Jouf University campus (Latitude: 29° 47′ 1.9" N, Longitude: 40° 2′ 42.2" E) located at Al-Jouf Province, Saudi Arabia. Several experimental data were conducted on the system on the aim of estimating the overall efficiency. All these data were conducted on a sunny day from eight at morning to four in the afternoon with a 15-minutes interval between each measurement of individual parameters (water temperatures, air temperatures, etc.). These measured parameters were used to estimate the overall efficiency of the constructed system. Figs. 3–6 show the measured solar intensity, wind velocity, water temperatures, and average chamber (air) temperature over the entire period of experiment. The solar intensity measurement showed a low value in the early morning then gradually increased to the peak value at noon time then gradually decreased to the lowest value before sunset, at end of the experiment. As solar energy is the main energy input to the system, the solar intensity trend will reflect on the hourly overall efficiency. The wind velocity measurement showed scattered values during the entire period of experiment (see Fig. 4). The wind velocities ranged from 0.5 to 2.5 m/s. The measured water inlet and out temperatures across the chamber control volume is shown in Fig. 5. Approximately, an average of 0.4°C temperature difference is observed between the inlet and out temperatures recorded at the same time. Another interesting observation is that the water temperature measurement shows a comparable trend with the solar intensity measurement, but with a lag of nearly an hour and a half between the two trends. The peak water temperature was recorded at 1:30 PM (see Fig. 5) while the peak solar intensity was recorded at 12:00 PM (see Fig. 3). It is speculated that this lag is due to the storage effect in the system. The average air temperature measurement inside the chamber is illustrated in Fig. 6. This average temperature is based on four thermocouples measurement (Fig. 2-f). The average air temperature showed a gradual increase until noon time then slightly decreased until end of

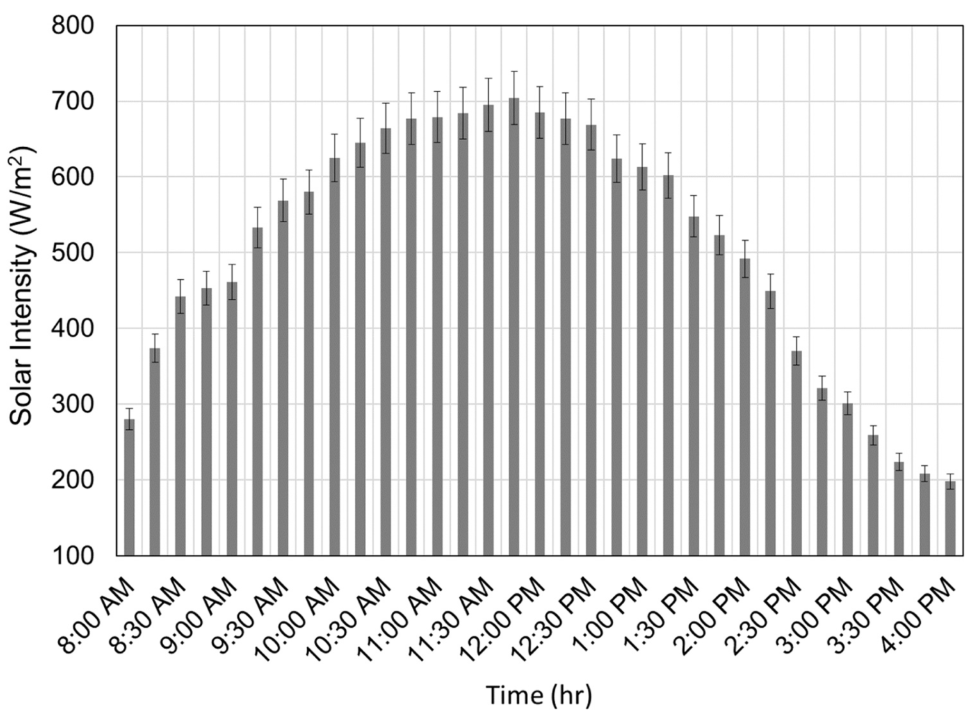

Fig. 3. Measured solar intensity during the experiment.

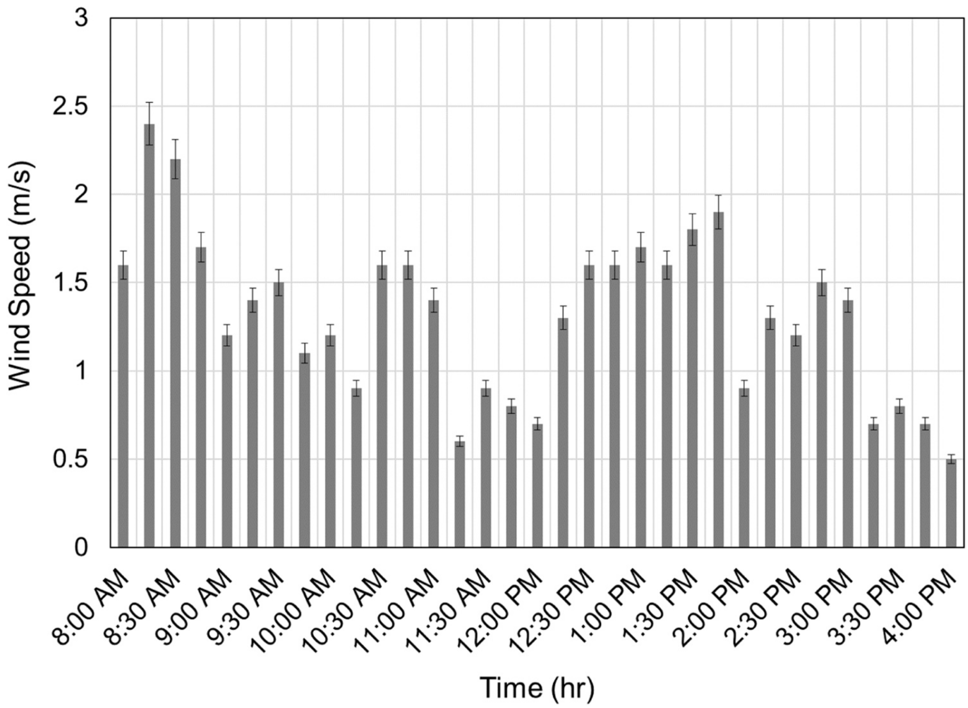

Fig. 4. Measured wind velocity during the experiment.

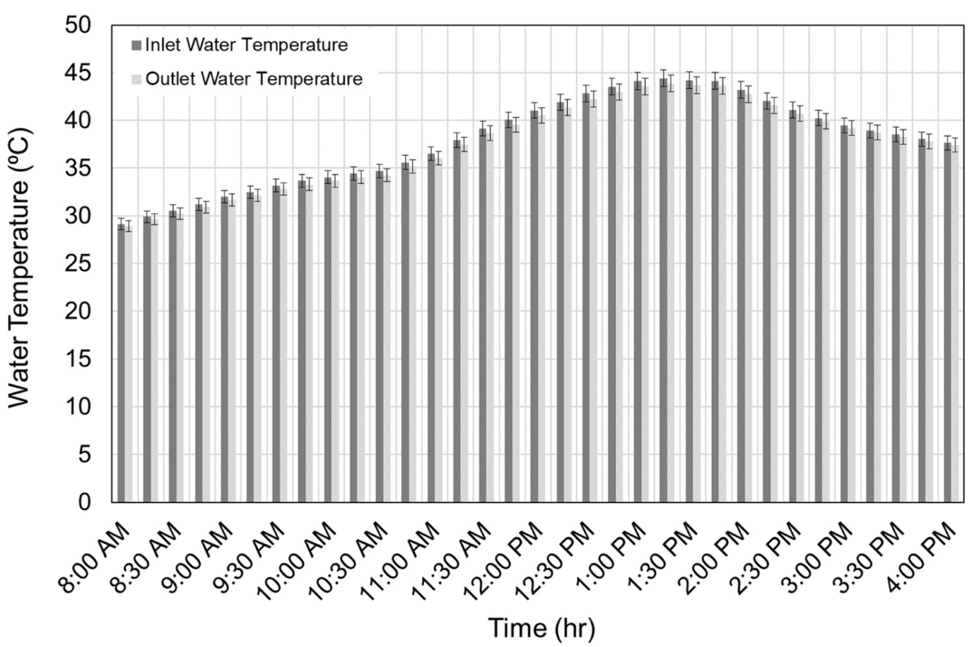

Fig. 5. Measured inlet and outlet water temperatures across the chamber during the experiment.

the experiment. Overall, this heating system led the air temperature inside the chamber to be increased from 15ºC (initial temperature) to 33ºC (final temperature) during the eight-hours experiment with a peak value of ~ 40ºC at 12:30 PM. However, these temperatures are way-off from the recommended values in ANSI/ASHRAE Standard 55 (ANSI/ASHRAE Standard 55, 2020) comfort zone (approximately 23ºC). The measured temperature values in the chamber is expected to be less in reality where actual residential buildings are not well insulated and controlled (i.e., no infiltration or other heat losses) as the chamber is constructed. In case of the heat utilized from solar energy was found to be more than the required energy for air heating, the surplus energy can be used for other domestic uses (i.e., cleaning or drying). Based on the individual measured parameters, the system overall efficiency was estimated according to Eq. (7), which was described in the previous section. The results of overall efficiency versus time along the experiment is shown in Fig. 7. This figure shows that the overall efficiency approximately varies between 35 % and 62 % with an average of 51 % over the eight-hours experiment. The initial overall efficiency (at 8:00 AM) is the lowest estimated value (approximately 35 %) because of the weak radiation from the sun at this time. The overall efficiency gradually increases to the maximum value (approximately 62 %) at 1:00 PM because of the strong radiation from the sun at noon time. The overall

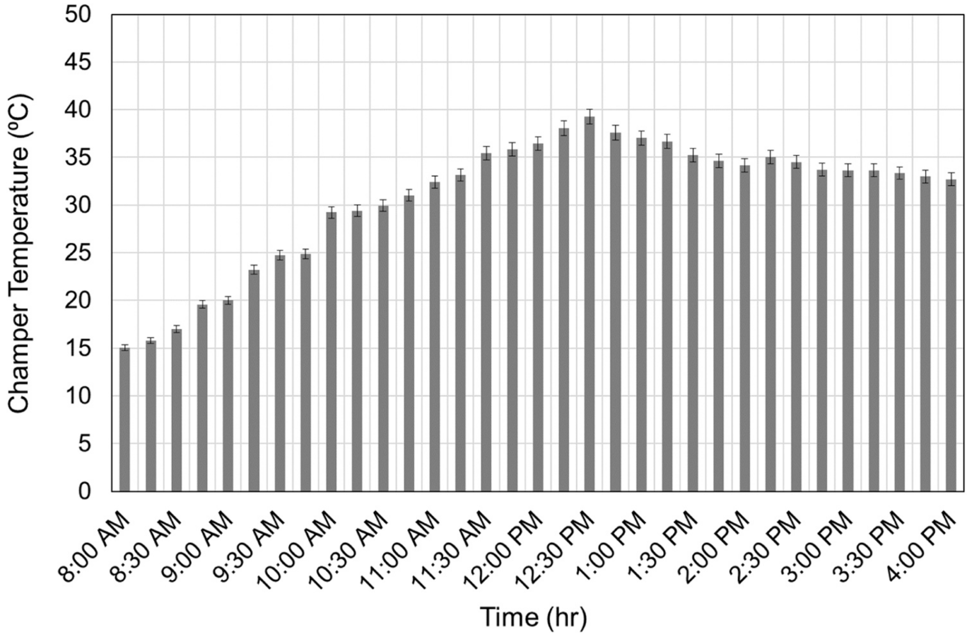

Fig. 6. Measured average air temperature inside the chamber during the experiment.

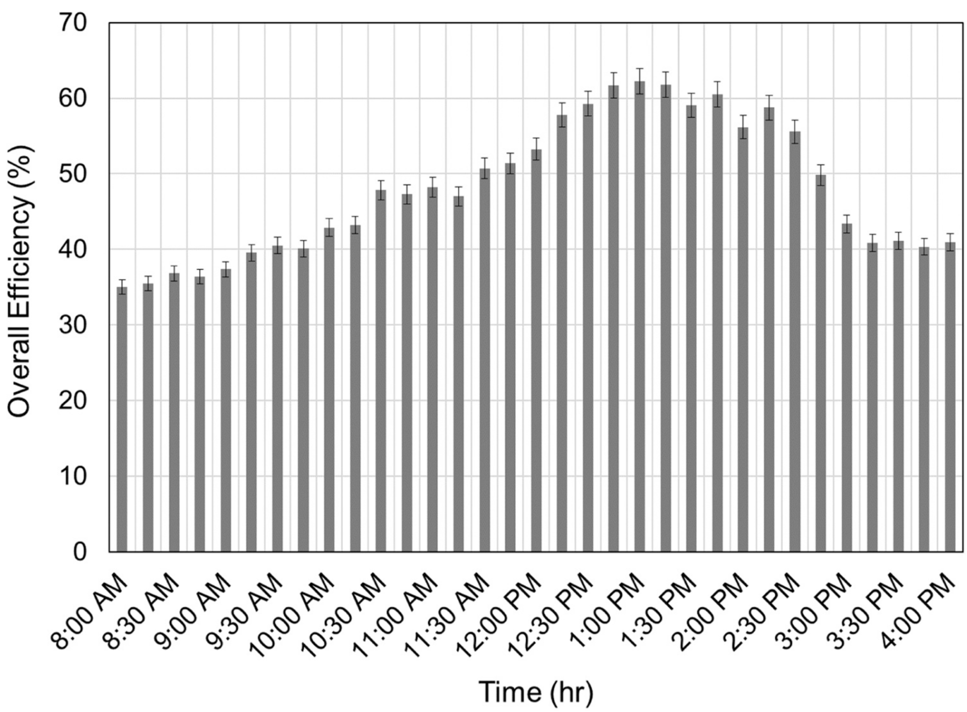

Fig. 7. Estimated overall efficiency of the constructed system during the experiment.

efficiency gradually decreases to approximately 40 % at end of the experiment as the radiation from the sun becomes lower. As discussed, it is observed that solar radiation is the important parameter that affects the performance of the constructed system, where the maximum efficiencies occur at the peak solar intensities and vice versa. The scattered wind velocity distribution during the experiment didn't show significant effect on the trend of system overall efficiency.

5. Cost analysis

In the design phase, cost of the solar heating system and its resulting benefits are important factors. Thus, a feasibility assessment of the

Table 3
Cost (I) of the constructed solar heating system.

Item	Description	Cost (SAR)
1	Flat plate solar collector and storage tank	2900
2	Chamber construction and installation	750
3	Two circulation pumps	900
4	Copper tube loop and flexible hose tube	150
5	Measuring instruments	1250
Total cost (I)		5950

proposed project should be implemented to compare the associated costs to the resulting benefits. A benefit-cost ratio (BCR) is an indicator that was used to evaluate the economic feasibility of the constructed solar heating system. The BCR is the ratio of benefits to costs where the project is economically feasible if the value of BCR is at least one. If the project's BCR is less than one, the project's costs outweigh the benefits and it should not be considered. The formula of BCR is expressed as (Ruegg and Marshall, 2013):

$$BCR = \frac{\text{Benefits of the Project}(B)}{\text{Initial Capital Cost}(I) \quad + \text{Operating \& Maintenance Costs}(O\&M)} \tag{8}$$

For estimating the BCR indicator of the present solar heating system, the benefits and costs are estimated as follow:

Benefits (B): Based on the experimental results that demonstrated an average energy gain (Q_u) of 0.792 (kW) obtained to heat the chamber during the experiment and the cost of residential electricity usage in Saudi Arabia of 0.18 (SAR/kWh), then the benefit from installing this solar heating system for 20 years (typical lifetime of a solar system for domestic water heating (Serban et al., 2016; Lin et al., 2015) will be

0.792 (kW) x 365 (day/year) x 24 (h/day) x 20 (year) x 0.18 (SAR/kWh) = 24977 SAR (9)

Cost (I): Table 3 lists the initial cost of individual items used to construct and test the solar heating system with a total cost (I) of 5950 SAR.

Costs (O & M): It is assumed that the constructed solar heating system has an operating cost only due to running the water pumps and a negligible maintenance cost. The cost of running the two water pumps is

0.37 (kW) x 365 (day/year) x 24 (h/day) x 20 (year) x 0.18 (SAR/kWh) = 11668 SAR (10)

Based on the benefits and costs estimated above, the BCR value for the constructed solar heating system is

$$BCR = \frac{24977}{5950 + 11668} = 1.42 \tag{11}$$

Thus, this estimated value of BCR indicates that the present solar heating system is economically feasible.

6. Summary and conclusions

A prototype of a solar energy system for heating a residential building was constructed and tested at Al-Jouf Province, Saudi Arabia (Latitude: 29° 47′ 1.9” N, Longitude: 40° 2′ 42.2” E) on January 2022. In this system, a flat plate solar collector was utilized to provide a heated water into the underfloor of a small-scale chamber, that represented the residential building, to heat the air inside the chamber on the aim of achieving thermal comfort in winter seasons. In order to evaluate the overall efficiency of the constructed system, several parameters were conducted during an eight-hours experiment. The experimental results led to an estimated minimum and maximum overall efficiency of 35 % and 62 % respectively, and an average overall efficiency of 51 % over the eight-hours experiment. It was observed from experimental results that the solar intensity is the important parameter that affects the overall efficiency distribution over the experiment duration, where the maximum efficiencies occurred at the peak solar intensities and vice versa. Additionally, a feasibility study was carried out to assess whether the constructed heating system is feasible or not. This study demonstrated that the proposed system is economically feasible where the estimated BCR value is 1.42. This present research can be extended to future work to: (1) investigate model size effect on the system energy efficiency and economic feasibility, and (2) apply theoretical methods to evaluate and validate such techniques.

CRediT authorship contribution statement

Waleed A. Abdelmaksoud: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: This work was funded by the Deanship of Graduate Studies and Scientific Research at Jouf University under grant No. (DGSSR-2023-02-02001).

Data availability

Data will be made available on request.

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